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1、波动方程：

$$\frac{\partial^2 u}{\partial t^2} = a^2 \nabla^2 u$$

于柱坐标下，

$$-\frac{1}{a^2}\frac{\partial^2 u}{\partial t^2} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

令 $u = T(t)V(r,\theta,z)$

代入，

$$-\frac{T''}{a^2 T} = \frac{\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial z^2}}{V} = -\lambda$$

$$\Rightarrow \begin{cases} T'' - a^2\lambda T = 0 \\ \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial z^2} + \lambda V = 0 \end{cases}$$

令 $V(r,\theta,z) = Z(z)\phi(r,\theta)$

代入，

$$\frac{\lambda\phi + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \phi}{\partial \theta^2}}{\phi} = \frac{-Z''}{Z} = -\mu$$

$$\Rightarrow \begin{cases} Z'' - \mu Z = 0 \\ \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \phi}{\partial \theta^2} + (\lambda+\mu)\phi = 0 \end{cases}$$

令 $\phi(r,\theta) = R(r)\Theta(\theta)$，

代入，

$$\frac{\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial R}{\partial r}\right) + \left[\frac{1}{r^2}\frac{\partial^2 \Theta}{\partial \theta^2} + (\lambda+\mu)\Theta\right]R}{-R} = \frac{\frac{1}{r^2}\frac{\partial^2 \Theta}{\partial \theta^2} + (\lambda+\mu)\Theta}{\Theta} = -k$$

$$\Rightarrow \begin{cases} \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial R}{\partial r}\right) - kR = 0 \\ \frac{1}{r^2}\frac{\partial^2 \Theta}{\partial \theta^2} + (\lambda+\mu+k)\Theta = 0 \end{cases}$$

综上，$u = T(t)Z(z)R(r)\Theta(\theta)$，

$$\begin{cases} \frac{1}{r}\frac{d}{dr}(r\frac{dR}{dr}) - kR = 0 \\ \frac{1}{r^2}\frac{d^2\Theta}{d\theta^2} + (\lambda+\mu+k)\Theta = 0 \\ T'' - a^2\lambda T = 0 \\ Z'' - \mu Z = 0 \end{cases}$$

同理对于球坐标，

$$-\frac{1}{a^2}\frac{d^2u}{dt^2} + \frac{1}{r^2}\frac{d}{dr}(r^2\frac{du}{dr}) + \frac{1}{r^2\sin\theta}\frac{d}{d\theta}(\sin\theta\frac{du}{d\theta}) + \frac{1}{r^2\sin^2\theta}\frac{d^2u}{d\phi^2} = 0$$

逐次分离变量得，

$$\begin{cases} T'' - a^2\lambda T = 0 \\ (\lambda+\mu)R + \frac{1}{r^2}\frac{d}{dr}(r^2\frac{dR}{dr}) = 0 \\ \frac{1}{r^2\sin^2\theta}\frac{d^2\Phi}{d\varphi^2} - (\mu+k)\Phi = 0 \\ \frac{1}{r^2\sin\theta}\frac{d}{d\theta}(\sin\theta\frac{d\Theta}{d\theta}) + k\Theta = 0 \end{cases}$$

A.5

2、(a)

$$W'' - z^2W = 0$$

$z=0$ 时，为常点，令 $W = \sum_{k=0}^{\infty} C_k z^k$，代入，

$$\sum_{k=0}^{\infty}\left[(k+4)(k+3)C_{k+4} + C_k\right] = 0$$

$$\Rightarrow C_{4k+4} = \frac{-1}{16(k+1)(k+\frac{7}{4})}C_{4k},\ C_2 = C_3 = 0$$

$$C_{4k} = \frac{3(-1)^k}{2^{4k+2}\Gamma(k)\ \Gamma(k-\frac{1}{4})} C_4 \quad , k=1,2,\cdots$$

同理,

$$C_{4(k+1)} = \frac{(-1)^k}{2^{4k}\Gamma(k)\Gamma(k-\frac{1}{4})} C_1 \quad , k=1,2,\cdots$$

三)

$$y_1 = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{4k}\Gamma(k)\Gamma(k-\frac{1}{4})} C_1 x^{4(k+1)} \quad , k=0,1,2,\cdots$$

$$y_2 = \sum_{k=1}^{\infty} \frac{3(-1)^k}{2^{4k+2}\Gamma(k)\Gamma(k-\frac{1}{4})} C_4 x^{4k} \quad , k=1,2,\cdots$$

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2、(C)

令 $w = \sum_{k=0}^{\infty} C_k x^k$, 代入,

$$(k-1)C_k - (k+2)C_{k+2} = 0$$

已?

$$C_{k+2} = \frac{k-1}{k+2} C_k \quad , \quad C_2 = 0,\ C_3 = 0$$

$\Rightarrow$) $C_k = 0,\ k = 1,2,\cdots$

EP:

$$\begin{cases} W_1(z) = C_0 \\ W_2(z) = 1, z^{-1} \end{cases}$$